

(PID 0) starts running. Here, we install a free-as-in-freedom boot-loader called GRUB, which is commonly referred to as the successor of the earlier LILO boot-loader. Install GRUB as follows:

```
[root@chroot /usr/src/linux]# cd
[root@chroot /]# emerge grub -av
```

Configure GRUB by creating (or modifying, in some cases) its configuration file, `/boot/grub/grub.conf`. GRUB has unique identifiers for disks and partitions; the following is an example with `sda1` as the root (`/`) partition, with no separate `/boot` partition:

```
[root@chroot /]# nano /boot/grub/grub.conf
#Which listing should boot as default, origins from zero. In this case, we
only have one listing called "Gentoo Linux"
default 0
#seconds to wait before booting the default
timeout 10

#Listing 0
title Gentoo Linux
root (hd0,0)
kernel /boot/bzImage root=/dev/sda1
```

Once done with the configuration file, install GRUB to the hard disk's MBR as follows:

```
[root@chroot /]# grub
GNU GRUB version 0.97 (640K lower / 3072K upper memory)
[ Minimal BASH-like line editing is supported. For the first word, TAB
lists possible command completions. Anywhere else TAB lists the possible
completions of a device/filename. ]
```

```
grub> root (hd0,0) #define the root device
<...>
grub> setup (hd0) #install GRUB to MBR
<...>
grub> quit
[root@chroot /]#
```

Finalising the installation

We are almost done; we need to modify `fstab` so that the correct disk partitions are mounted on booting up:

```
[root@chroot /]# nano /etc/fstab

# /etc/fstab: static file system information.
#
# <fs>          <mountpoint>    <type>          <opts>
<dump/pass>
# <file system> <mount point> <type> <options> <dump> <pass>
/dev/sda1      /                ext3 defaults 1      2
```

```
/dev/sda2      none            swap sw        0      0

# glibc 2.2 and above expects tapfs to be mounted at /dev/shm for
# POSIX shared memory (shm_open, shm_unlink).
# (tapfs is a dynamically expandable/shrinkable ramdisk, and will
# use almost no memory if not populated with files)
shm            /dev/shm       tapfs          nodev,nosuid,noexec 0 0
```

Finally, we install the necessary tools like the DHCP client, system logger and cron:

```
[root@chroot /]# emerge net-misc/dhclient sys-process/vixie-cron app-admin/
syslog-ng -av
```

Add `cron` to the default run-level, and the system logger to the boot run-level (make them run upon boot, but run system logger before `cron`):

```
[root@chroot /]# rc-update add vixie-cron default
* vixie-cron added to runlevel default
[root@chroot /]# rc-update add syslog-ng boot
* syslog-ng added to runlevel boot
```

For network configurations, refer to `/etc/conf.d/net.example`. To use DHCP to obtain address information for the Ethernet connection `eth0`, issue the following command when the network starts up on boot:

```
[root@chroot /]# nano /etc/conf.d/net
config_eth0=( "dhcp" )
[root@chroot /]# rc-update add net.eth0 default
```

Set the root password:

```
[root@chroot /]# passwd
```

Last, we need to create user accounts and set passwords:

```
[root@chroot /]# useradd -c "John Smith" john
[root@chroot /]# passwd john
New password:
Retype new password:
passwd: password updated successfully
```

Now we're all done! To reboot to the new system, exit out of the `chroot` environment, unmount the file-systems and restart the computer, making sure that the hard disk comes first in the boot order by changing the BIOS settings to do so, or by removing the CD or USB stick that you installed from.

```
[root@chroot /]# exit
[lived /]# umount /mnt/gentoo/proc /mnt/gentoo/dev /mnt/gentoo
[lived /]# reboot
```